

How Low Can Chronos-2 Go? Post-Training Quantization of a Time-Series Foundation Model

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Abstract

Time-series foundation models (TSFMs) such as Chronos-2 deliver strong zero-shot probabilistic forecasts, but their deployment cost is rarely studied and, unlike large language models, *nothing* is known about how they respond to post-training quantization (PTQ). We present the first systematic PTQ study of a TSFM: eighteen weight-quantization configurations spanning six method families (round-to-nearest, torchao, bitsandbytes INT8/NF4, HQQ, GPTQ, and activation-aware scale folding) and two- to eight-bit precision, evaluated zero-shot on the 27-dataset Chronos benchmark. Beyond point (MASE) and probabilistic (weighted quantile loss, WQL) accuracy, we introduce two calibration-aware diagnostics—mean absolute coverage error and a quantile-crossing rate—because a probabilistic forecaster can degrade in ways aggregate accuracy cannot see. Four findings emerge. (i) 8-bit quantization is free for every method ($\leq 1.3\%$ WQL change) and shrinks the model $4\times$. (ii) Hessian-aware GPTQ retains full-precision accuracy down to **3 bits** (WQL retention 1.05, 95% CI [0.99, 1.14]), where the naive baseline degrades 70%. (iii) The *method* matters far more than the *bit-width*: at 4 bits, retention spans 0.996–1.66 across methods. (iv) Interval calibration and quantile monotonicity degrade *before* accuracy does—8-bit quantization leaves WQL untouched yet quadruples the quantile-crossing rate—so accuracy-only evaluation overstates the quality of a quantized forecaster. We release the framework, which reproduces published per-task benchmark numbers to $< 10^{-6}$, and all per-series predictions.

1 Introduction

Time-series foundation models (TSFMs)—Chronos [1], Chronos-2 [2], TimesFM, Moirai—forecast unseen series zero-shot and are increasingly deployed on CPU-only servers, edge devices, and high-throughput batch pipelines where full-precision inference is costly. Quantization is the standard efficiency lever for large language models (LLMs), but its lessons do not obviously transfer to TSFMs, which are small (Chronos-2 has 120M parameters, a regime where quantization bites harder [9]), encoder-only and non-autoregressive (no KV cache, so LLM decode-centric methods do not apply), and—critically—emit *probabilistic* forecasts whose calibration and quantile monotonicity matter, not just point accuracy.

Despite an extensive LLM-quantization literature, **no published work quantizes the weights of any Chronos-family or comparable TSFM** (in that literature “quantization” usually denotes Chronos-1’s input value-*binning*, a distinct notion). We close that gap with a controlled study across method families and bit-widths, and argue that evaluating a *quantized*

probabilistic forecaster demands metrics beyond those on TSFM leaderboards.

Contributions.

- The first systematic PTQ study of a TSFM: 18 configurations across 6 method families and 5 bit-widths, zero-shot on 27 datasets, with bootstrap confidence intervals on every aggregate.
- A calibration-aware evaluation protocol for compressed probabilistic forecasters, adding interval-coverage error and a quantile-crossing rate, which reveal failure modes invisible to WQL/MASE.
- Findings and a deployment recommendation: GPTQ reaches full-precision accuracy *and* calibration at 3–4 bits, whereas simpler methods trade calibration first.
- An open, reproducible framework that reproduces published benchmark numbers to floating-point precision and releases all per-series predictions.

2 Setup

Model. Chronos-2 [2] (amazon/chronos-2) is a 119.5M-parameter, T5-derived *encoder-only* transformer (12 layers, $d_{\text{model}}=768$, RoPE, alternating “time” and “group” attention). It ingests patches of a per-instance-normalized series through a residual embedding (no input tokenization) and emits, in one non-autoregressive forward pass, direct multi-step forecasts at 21 quantile levels. It ships in fp32 (478 MB). All quantization is applied to the model’s nn.Linear weights; LayerNorm, the input embedding, and the quantile head follow standard practice and stay in fp32, and bf16 (not fp16, which overflows in T5-lineage models) is the deployment baseline.

Methods. We evaluate weight-only PTQ across six families runnable on commodity hardware: **RTN** (symmetric round-to-nearest, per-channel/ per-tensor, plus a simulated sub-8-bit sweep for the accuracy-bit curve); **torchao** INT8 weight-only and INT8 dynamic-activation (W8A8); **bitsandbytes** LLM.int8() [4] and NF4 [5]; **HQQ** [3], a calibration-free half-quadratic solver, at 8/4/3 bits; **GPTQ** [7], a Hessian-aware error-compensating quantizer we implement directly on the encoder’s linear layers with one-shot calibration; and **AWQ-style** activation-aware scale folding [8]. Methods are exercised at W8, W4 (group size 64/128), W3, and W2.

Protocol & metrics. We evaluate zero-shot on the 27-dataset Chronos benchmark [1] (190,674 forecasts) using the `fev` library—the harness the Chronos-2 authors use—so our numbers are directly comparable with the public leaderboard. Point accuracy is MASE and probabilistic accuracy

Table 1: Representative variants, full 27-task benchmark. Retention = geometric-mean WQL(quant)/WQL(fp32), [95% bootstrap CI]. $c80$ = empirical coverage of the nominal-80% interval (fp32: 0.750). QCR = quantile-crossing rate (fp32: 0.039). BPW = effective bits/weight incl. scales; † accuracy-exact simulation (dequantized grid in fp32)—BPW is the *achievable* footprint.

Variant	Bits	BPW	WQL ret. [CI]	MASE	c80	QCR
fp32 (ref)	32	32.0	1.000	1.000	.750	.039
RTN W8	8	8.03	0.999 [0.997,1.00]	1.001	.749	.149
GPTQ W8	8	8.13†	1.000 [0.990,1.01]	1.003	.749	.064
HQQ W8	8	9.00	1.002 [1.00,1.00]	1.000	.746	.086
GPTQ W4	4	4.13†	1.002 [0.985,1.02]	1.011	.756	.254
HQQ W4	4	5.00	1.060 [1.02,1.12]	1.027	.630	.420
NF4 W4	4	4.13	1.065 [1.00,1.15]	1.026	.686	.396
AWQ W4	4	4.03†	1.639 [1.38,2.01]	1.409	.481	.599
GPTQ W3	3	3.25†	1.047 [0.988,1.14]	1.059	.791	.366
HQQ W3	3	4.20	1.229 [1.09,1.48]	1.143	.666	.589
RTN W2	2	2.0†	4.082 [3.08,5.57]	3.108	.060	.523

is weighted quantile loss (WQL, 9 levels); per task each is normalized by a seasonal-naive baseline and aggregated by geometric mean into a *skill score*, with pairwise *win rates*. Our full-precision Chronos-2 attains 0.425 WQL skill and an 88% win rate over seasonal-naive, reproducing the paper’s ordering; as a reproduction gate, our seasonal-naive matches the published per-task MASE/WQL to $< 10^{-6}$.

For quantized models the key quantity is **retention**: the geometric-mean ratio WQL(quant)/WQL(fp32) over the 27 tasks (1.00 = parity), with a 1000-resample bootstrap 95% CI. Because a probabilistic forecaster can lose quality without moving WQL, we add two diagnostics. **MACE** (mean absolute coverage error) averages $|\text{empirical} - \text{nominal}|$ over the central 80/60/40/20% intervals. **QCR** (quantile-crossing rate) is the fraction of forecast points whose predicted quantiles are non-monotone—a structural defect a direct quantile head can acquire under quantization. All per-series predictions are persisted so any further metric needs no model re-run.

3 Results

Table 1 summarizes representative configurations; Fig. 1 plots the full accuracy-bit curve and calibration. The complete 18-variant matrix is released.

F1—8-bit is free, for every method. All six W8 configurations retain WQL within [0.999, 1.013] and MASE within [1.000, 1.011] of fp32 (Table 1; Fig. 1a), at $4\times$ compression. Even W8A8 with dynamic-activation quantization costs only 1.3%. Eight-bit quantization of Chronos-2 is, for practical purposes, lossless regardless of method.

F2—GPTQ reaches full-precision accuracy at 3 bits. GPTQ retains parity at 4 bits (retention 1.00, both group sizes; CIs include 1.0) and, remarkably, at **3 bits** (1.047, CI [0.988, 1.137]—statistically indistinguishable from fp32), where HQQ loses 23% and naive RTN 70% (Fig. 1a). Hessian-aware error compensation moves the accuracy cliff at least one bit lower than any calibration-free method. (A dev-subset signal that GPTQ *beat* fp32 did not survive the full 27-task

Table 2: Inference efficiency (storage-real variants, NVIDIA RTX 2000 Ada laptop, context 2048, horizon 64). Peak mem and latency at batch 1; throughput at batch 256.

Model	Size MB	Peak GPU MB (b1)	GPU lat. ms (b1)	Thr. ser./s
fp32	478	493	30	110
INT8 (HQQ)	120	176	42	122
NF4 (W4)	60	87	45	120

evaluation, resolving to parity—we report the confirmed result.)

F3—method dominates bit-width. At a fixed 4 bits, WQL retention spans 0.996 (GPTQ) to 1.66 (RTN)—a 66-point gap—whereas moving a *good* method from 8 to 4 bits costs $\leq 6\%$ (Fig. 1a). For a TSFM, *how* you quantize matters far more than *how much*, inverting the intuition that bit-width is the primary knob.

F4—calibration and monotonicity degrade before accuracy, and only GPTQ preserves them. This is the study’s central cautionary finding. At 8 bits, accuracy is perfectly retained yet the quantile-crossing rate already quadruples (0.039 $\rightarrow$ 0.149 for RTN) —a defect in the quantile head that WQL and MASE cannot see. At 4 bits, GPTQ keeps interval coverage intact (c80 0.756 vs fp32 0.750; Fig. 1b) while HQQ and NF4 sag to 0.630 and 0.686—a user trusting the “80%” band of HQQ-W4 actually holds a 63% band. GPTQ is the *only* 4-bit method preserving both accuracy and calibration; no method fully preserves quantile monotonicity. Evaluating a quantized probabilistic forecaster on accuracy alone therefore overstates its deployable quality [6].

F5—damage is tail-concentrated. Mean retention hides worst cases. HQQ-W3 averages 1.23 but its worst task is $8.2\times$ and 44% of tasks exceed 1.10; GPTQ-W4 has a median per-task ratio of 0.998 and only 4% of tasks above 1.10. Per-series distributions—released for every run—are the honest picture, echoing the “hard inputs degrade first” pattern from encoder-model quantization.

F6—activation-aware folding is a rigorous negative result. For Chronos-2 the AWQ scale fold is *exact* (pre-norm RMSNorm weights absorb the scale; the ReLU MLP is positively homogeneous; $v \rightarrow o$ folds cleanly), which we verify numerically. Yet at 4 bits AWQ-fold+RTN (1.64) barely differs from plain RTN (1.66) and is far behind GPTQ (Fig. 1a). On this model, error *compensation* (GPTQ) is the effective mechanism; activation *scaling* alone is not.

Efficiency. Storage-real 4-bit models occupy ≈ 60 MB ($8\times$ smaller) and cut peak batch-1 GPU memory from 493 MB (fp32) to 87 MB for NF4 ($5.7\times$; Table 2). Throughput is flat (~ 120 series/s at batch 256, vs 110 for fp32): our reference dequantize-then-GEMM paths buy footprint, not speed—the expected outcome for unfused weight-only kernels, and why optimized-kernel or activation-quantized paths are needed for latency. Memory-bound and edge deployments benefit immediately.

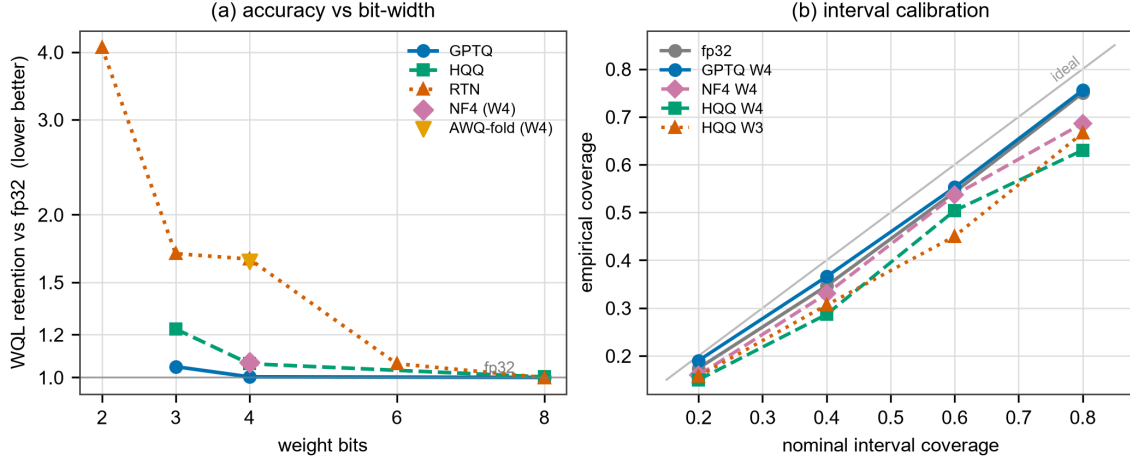


Figure 1: (a) WQL retention vs weight bits by method (log scale; lower is better, 1.0 = fp32). GPTQ tracks fp32 to 3 bits; the method spread at 4 bits exceeds the 8→4-bit gap of any single method; AWQ-fold sits on the RTN line. (b) Empirical vs nominal interval coverage: GPTQ-W4 tracks fp32 while HQQ/NF4 systematically under-cover.

4 Discussion, Limitations, Conclusion

Recommendation. To deploy Chronos-2: use 8-bit for a free $4\times$ reduction with any toolkit; for 4 bits and below, use GPTQ (or another Hessian-aware method) rather than a calibration-free quantizer—not merely for accuracy but because it is the only method that preserves interval calibration, the property forecasting users depend on. Report calibration and quantile-crossing alongside WQL whenever a probabilistic model is compressed.

Limitations. We study one 120M model on univariate zero-shot tasks; covariate/multivariate tasks, other TSFM sizes, and quantization-aware training are future work. Our GPTQ/AWQ arms are accuracy-exact simulations (dequantized grids in fp32 containers); realized latency for them requires packed kernels we do not implement (achievable footprint is reported). Latency is hardware-specific; retention ratios are the transferable claim. We assess but exclude KV-cache/vector methods such as TurboQuant [10], which do not apply to a cache-free encoder.

Conclusion. Chronos-2 quantizes remarkably well: 8-bit is free, and Hessian-aware 4- and 3-bit quantization retains full-precision forecasting accuracy. But probabilistic structure—interval calibration and quantile monotonicity—degrades before accuracy, and only error-compensating methods preserve it. Compressing a probabilistic forecaster is therefore not merely an accuracy-retention problem, and should not be evaluated as one. Code, configs, per-task summaries, and all per-series predictions are released; every number is regenerated by one command.

References

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